

Variance Reduction Note 3: The Convergence of PAGE

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Abstract

The probabilistic gradient estimator (PAGE) alternates randomly between an accurate gradient refresh and a cheap recursive correction. This third expository note on recursive variance reduction explains the algorithm and gives a self-contained convergence proof for smooth nonconvex optimization. A Lyapunov function combines objective decrease with estimation error, turning the refresh probability into an explicit contraction parameter. The resulting expected oracle complexity is $O(n + \sqrt{n}\varepsilon^{-2})$ for finite sums and $O(\varepsilon^{-3})$ for online problems, with the other problem parameters fixed. Initialization, refreshes, and both evaluations in each gradient difference are counted. The presentation is pedagogical and makes no claim of a new algorithm or convergence rate.

1 Random refreshes and recursive corrections

Consider the smooth, possibly nonconvex objective

$$\min_{x \in \mathbb{R}^d} F(x) := \mathbb{E}_\xi[f(x; \xi)], \quad g(x; \xi) := \nabla f(x; \xi). \quad (1)$$

Recursive estimators such as SARAH and SPIDER [23, 26] use shared-sample differences to track changes in the gradient. A refresh prevents the accumulated tracking error from becoming too large. PAGE, introduced by Li et al. [31], performs this refresh with a specified probability at every iteration. The resulting random refresh times remove the need for a prescribed epoch length.

The method has two computational scales: an expensive refresh batch and a smaller batch of gradient differences. The refresh probability balances their expected costs. Its mathematical role is equally direct: refreshing replaces the old estimation error, so that error survives with probability $1 - p$. This yields a short Lyapunov proof with a constant step size. Related probabilistic designs include loopless SVRG [18]; STORM instead incorporates a damped recursive correction at every step [33]. For a general introduction to variance reduction, see [2].

2 The oracle and the PAGE update

Assumption 1. *The differentiable function F is bounded below by a finite number $F_\star$. For some $L > 0$ and all $x, y \in \mathbb{R}^d$,*

$$\mathbb{E}[g(x; \xi)] = \nabla F(x), \quad \mathbb{E} \|g(x; \xi) - g(y; \xi)\|^2 \leq L^2 \|x - y\|^2. \quad (2)$$

For online problems, additionally assume, for some $\sigma \geq 0$,

$$\mathbb{E} \|g(x; \xi) - \nabla F(x)\|^2 \leq \sigma^2. \quad (3)$$

Fresh samples are independent of the past and of one another. A sampled function can be evaluated at two specified points. Norms are Euclidean.

Jensen's inequality implies that F is L -smooth. For a finite sum $F(x) = n^{-1} \sum_{i=1}^n f_i(x)$, use uniform indices and $g(x; i) = \nabla f_i(x)$. Exact full-gradient refreshes then eliminate the need for (3).

Choose a fixed step size $\eta > 0$, a probability $p \in (0, 1]$, and integer batch sizes $B, b \geq 1$. Initialize v_0 with a refresh at a deterministic x_0 . For $t = 0, \dots, T-1$, compute

$$x_{t+1} = x_t - \eta v_t, \quad (4)$$

then draw a fresh independent Bernoulli variable C_{t+1} with $\Pr(C_{t+1} = 1) = p$, and set

$$v_{t+1} = \begin{cases} \frac{1}{B} \sum_{i=1}^B g(x_{t+1}; \zeta_{t+1,i}), & C_{t+1} = 1, \\ v_t + \frac{1}{b} \sum_{i=1}^b [g(x_{t+1}; \xi_{t+1,i}) - g(x_t; \xi_{t+1,i})], & C_{t+1} = 0. \end{cases} \quad (5)$$

Both evaluations in each difference use the same sample. For finite sums, replace every refresh, including initialization, by the full gradient; set $B = n$ for cost accounting and sample inner indices independently with replacement. Return x_R for an independent uniform $R \in \{0, \dots, T-1\}$. The expected cost of this implementation is

$$\mathbb{E}N = B + T[pB + 2(1-p)b]. \quad (6)$$

This count includes the final estimator update, which can be omitted without changing the returned point or the guarantee.

3 A short convergence proof

Write $e_t = v_t - \nabla F(x_t)$, $Q_t = \mathbb{E} \|e_t\|^2$, $D_t = \mathbb{E} \|v_t\|^2$, and $G_t = \mathbb{E} \|\nabla F(x_t)\|^2$. Let $\nu^2 = \sigma^2/B$ online and $\nu^2 = 0$ for exact finite-sum refreshes. In both settings $Q_0 \leq \nu^2$.

Lemma 1 (Tracking error). *Under Assumption 1,*

$$Q_{t+1} \leq (1-p)Q_t + \frac{(1-p)L^2\eta^2}{b}D_t + p\nu^2. \quad (7)$$

Proof. Condition on all randomness used to construct x_t, v_t ; then x_{t+1} is also fixed. A refresh has conditional mean-square error at most ν^2 . In the recursive branch, write $e_{t+1} = e_t + Z_{t+1}$, where

$$Z_{t+1} = \frac{1}{b} \sum_{i=1}^b [g(x_{t+1}; \xi_i) - g(x_t; \xi_i)] - [\nabla F(x_{t+1}) - \nabla F(x_t)].$$

The increment has conditional mean zero. Independence within the batch, the variance identity, and (2) bound its conditional second moment by $L^2 \|x_{t+1} - x_t\|^2 / b$. Its cross term with e_t vanishes. Average over the fresh Bernoulli choice and the past, and use (4). $\square$

Lemma 2 (Descent). *For the update (4),*

$$\mathbb{E}F(x_{t+1}) \leq \mathbb{E}F(x_t) - \frac{\eta}{2}G_t - \frac{\eta}{2}(1-L\eta)D_t + \frac{\eta}{2}Q_t. \quad (8)$$

Proof. Apply L -smoothness to $x_{t+1} = x_t - \eta v_t$ and substitute

$$2 \langle \nabla F(x_t), v_t \rangle = \|\nabla F(x_t)\|^2 + \|v_t\|^2 - \|e_t\|^2.$$

Taking expectations gives (8); conditional unbiasedness of v_t is not required. $\square$

Theorem 1 (Expected stationarity). *Let $\Delta = F(x_0) - F_\star$ and $c = (1 - p)/(pb)$. If*

$$0 < \eta \leq \frac{1}{L(1 + \sqrt{c})}, \quad (9)$$

then PAGE satisfies

$$\mathbb{E} \|\nabla F(x_R)\|^2 \leq \frac{2\Delta}{\eta T} + \frac{Q_0}{pT} + \nu^2 \leq \frac{2\Delta}{\eta T} + \frac{\nu^2}{pT} + \nu^2. \quad (10)$$

Proof. Use the nonnegative Lyapunov quantity $\Phi_t = \mathbb{E}[F(x_t) - F_\star] + \eta Q_t/(2p)$. Multiply (7) by $\eta/(2p)$ and add (8). The Q_t terms cancel, giving

$$\Phi_{t+1} - \Phi_t \leq -\frac{\eta}{2}G_t - \frac{\eta}{2}[1 - L\eta - cL^2\eta^2]D_t + \frac{\eta}{2}\nu^2.$$

The bracket is nonnegative: with $r = \sqrt{c}$, condition (9) gives

$$L\eta + cL^2\eta^2 \leq \frac{1}{1+r} + \frac{r^2}{(1+r)^2} \leq 1.$$

Sum over $t = 0, \dots, T-1$ and use $\Phi_T \geq 0$ and $\Phi_0 = \Delta + \eta Q_0/(2p)$. Division by $\eta T/2$ yields (10), because R is independent and uniform. $\square$

Finite-sum complexity. Take exact refreshes, $B = n$, $b = \lceil \sqrt{n} \rceil$, $p = b/(n+b)$, and equality in (9). Then $c = n/b^2 \leq 1$, so $\eta \geq 1/(2L)$, and $Q_0 = \nu^2 = 0$. For a target $\varepsilon > 0$, choosing $T = \max\{1, \lceil 4L\Delta/\varepsilon^2 \rceil\}$ gives $\mathbb{E} \|\nabla F(x_R)\|^2 \leq \varepsilon^2$. Moreover, (6) gives

$$\mathbb{E}N = n + \frac{3nb}{n+b}T \leq n + 3bT = O\left(n + \frac{\sqrt{n}L\Delta}{\varepsilon^2}\right). \quad (11)$$

The $O(\sqrt{n})$ cost from rounding T is absorbed by n .

Online complexity. Choose

$$B = \max\{1, \lceil 2\sigma^2/\varepsilon^2 \rceil\}, \quad b = \lceil \sqrt{B} \rceil, \quad p = \frac{b}{B+b}, \quad (12)$$

and equality in (9). Again $c = B/b^2 \leq 1$, while $\nu^2 \leq \varepsilon^2/2$. With

$$T = \left\lceil \frac{4\Delta}{\eta\varepsilon^2} + \frac{1}{p} \right\rceil, \quad (13)$$

the first two terms in (10) sum to at most $\varepsilon^2/2$. Thus $\mathbb{E} \|\nabla F(x_R)\|^2 \leq \varepsilon^2$. Since $b/p = B+b$ and $\eta^{-1} \leq 2L$,

$$\begin{aligned} \mathbb{E}N &\leq B + 3bT = O\left(B + \frac{\sqrt{B}L\Delta}{\varepsilon^2}\right) \\ &= O\left(1 + \frac{\sigma^2}{\varepsilon^2} + \left(1 + \frac{\sigma}{\varepsilon}\right) \frac{L\Delta}{\varepsilon^2}\right). \end{aligned} \quad (14)$$

For fixed problem parameters this is $O(\varepsilon^{-3})$ as $\varepsilon \downarrow 0$. In either setting, Cauchy–Schwarz also gives $\mathbb{E} \|\nabla F(x_R)\| \leq \varepsilon$.

Interpretation and scope. The term $Q_0/(pT)$ shows why initialization belongs in the proof. A sampled refresh leaves a noise floor ν^2 , which must decrease with the requested accuracy. The probability $p = b/(B + b)$ and $b \asymp \sqrt{B}$ balance the expected work while permitting a step size of order $1/L$. Cost bounds are in expectation over the random refreshes. The chosen horizons use Δ or an upper bound on it; online batch selection also uses a variance bound. The theorem concerns a random iterate and first-order stationarity. Linear rates under a Polyak–Łojasiewicz condition need an additional assumption and are treated in the original PAGE paper [31].

4 Recommended reading

Variance-reduction methods. Basic guarantees [1–3]; gradient memories and control variates [4–10]; sampling and acceleration [11–19]; nonconvex and recursive estimators [20–32]; momentum and adaptive schedules [33–43]; second-order extensions [44–47]; lower bounds [48].

Compositional optimization. Two-level approximation [49–52]; finite-sum and recursive estimators [53–56]; multilevel tracking [57–60]; coupled and partially observed blocks [61–65]; projection-free methods [66–70]; statistical-model applications and federated extensions [71–74].

Sign-based optimization. Quantization, signSGD, majority voting, and error feedback [75–79]; variance-reduced sign updates and Lion [80–82]; compressed gradient tracking [83–87]. Related distributed control-variate methods include SCAFFOLD, FedPAGE, decentralized hybrid estimators, and ProxSkip [88–91].

Minimax optimization. Finite-sum saddle-point methods, matrix games, and variance-reduced extragradient [92–95]; recursive, hybrid, adaptive, and decentralized nonconvex minimax methods [96–99]; a variational-inequality treatment [100].

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